# AIMO Proof Pilot — Final Grade

**Problem (#_ID):** 3_SINDEX **Submission:** model_6

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**Scoring scale (0/1/6/7) — two binary decisions:**

**Step 1. Is the solution essentially complete?**

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

## 1. Final mark

**Final mark** (choose one): $\boxed{6}$

## 2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

An essentially complete solution matching the official approach. It sets up the polynomials, proves the key lemma, correctly rules out index $\geq 6$, attains index 5 (so $N = 5$), and establishes $M = 16$ with a valid example. The one genuine defect is in the final bounding step: passing from $q_1 > 0$ to $q_1 \geq 1$ needs $q_1$ to be an integer — i.e. that $f/L_5$ has integer coefficients — which is used but never justified (`ME2`). Essentially complete with a minor error, so scores **6**.